Separating MAP- and CO_2-Associated Contributions to Cerebral Blood Flow Velocity

A multivariable transfer function analysis in mild cognitive impairment

Amogh Naganand

July 21, 2026

Core question: Does accounting for spontaneous CO_2 change how we interpret MAP–CBFV dynamics in NC and MCI?

Why this question matters

Pressure

MAP fluctuations are the standard input used to assess dynamic cerebral autoregulation.

Carbon dioxide

CO_2 is vasoactive and changes cerebral vascular tone in overlapping frequency ranges.

Cerebral output

Transcranial Doppler CBFV reflects the combined behavior of pressure, CO_2, and unmeasured mechanisms.

$$\text{MAP} + \text{CO}_2 \longrightarrow \text{CBFV}$$

Takeaway: A MAP-only model may mix pressure-associated dynamics with variation shared with CO_2.

SISO and MISO answer different questions

Separate SISO models

$$H_{\mathrm{MAP}}^{\mathrm{SISO}} = \frac{S_{\mathrm{MAP,CBFV}}}{S_{\mathrm{MAP,MAP}}}$$

$$H_{CO_2}^{\mathrm{SISO}} = \frac{S_{CO_2,\mathrm{CBFV}}}{S_{CO_2,CO_2}}$$

Pairwise input–output relationships

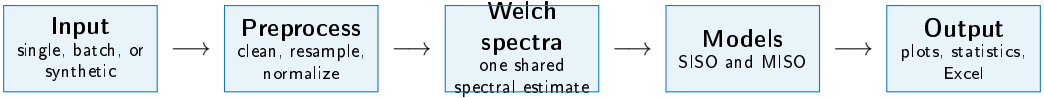
One simultaneous MISO model

$$\begin{bmatrix} S_{\mathrm{MAP,MAP}} & S_{\mathrm{MAP},CO_2} \\ S_{CO_2,\mathrm{MAP}} & S_{CO_2,CO_2} \end{bmatrix} \begin{bmatrix} H_{\mathrm{MAP}}^{\mathrm{MISO}} \\ H_{CO_2}^{\mathrm{MISO}} \end{bmatrix} = \begin{bmatrix} S_{\mathrm{MAP,CBFV}} \\ S_{CO_2,\mathrm{CBFV}} \end{bmatrix}$$

Each pathway is estimated while accounting for the other measured input

Central hypothesis: Simultaneously modeling MAP and CO_2 will produce different pathway-specific transfer function estimates than modeling each input separately. These conditional estimates may also reveal NC–MCI differences.

Pipeline



Consistent analysis path

The same preprocessing and Welch spectra are passed into both models. Subject results are retained for group means, variability, paired model comparisons, and independent NC–MCI comparisons.

Takeaway: Model differences reflect the transfer-function formulation, rather than separate preprocessing or spectral estimates.

Dataset and analysis settings

Current cohort

- **23 NC** participants
- **52 MCI** participants
- All 75 completed the current TFA workflow
- Similar age, sex, MAP, $P_{\mathrm{ET}}CO_2$, CBFV, and recording duration
- MoCA was lower in MCI, as expected

Core settings

- Resampled to 4 Hz
- CBFV normalized to percent baseline
- 128-s Hann windows, 50% overlap
- Minimum of 3 Welch windows
- Frequency range: 0–0.35 Hz
- Bands: VVLF, VLF, LF, HF
- Same smoothing kernel for every spectrum

Takeaway: With these settings, three Welch windows require at least 1,024 samples, or 256 seconds at 4 Hz.

NC signal and spectral overview

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`nc_groupoverview.png`

MCI signal and spectral overview

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`mci_groupoverview.png`

Reading a transfer-function result

Gain

$$|H(f)|$$

Magnitude of the frequency-specific CBFV response associated with an input oscillation.

Phase

$$\arg[H(f)]$$

Timing relationship between the input and CBFV. Group phase uses circular means and circular SD.

Coherence

Strength of the linear frequency-specific relationship.

SISO uses ordinary coherence; MISO reports pathway-specific partial coherence and combined multiple coherence.

Takeaway: The complete transfer function is frequency-dependent; band averages summarize these curves but do not replace them.

NC: MISO MAP-to-CBFV transfer function

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`nc_groupmisomaptocbv.png`

MCI: MISO MAP-to-CBFV transfer function

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`mci_groupmisomaptocbv.png`

NC: MISO CO_2-to-CBFV transfer function

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`nc_groupmisoco2tocbv.png`

MCI: MISO CO_2-to-CBFV transfer function

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`mci_groupmisoco2tocbv.png`

Comparing the Models

MISO versus SISO

For each participant and result:

$$d = \text{MISO value} - \text{SISO value}$$

$$H_0 : \text{ average difference} = 0.$$

NC versus MCI

For each MISO result, participant values in NC are compared with participant values in MCI.

$$H_0 : \ \overline{x}_{\mathrm{NC}} = \overline{x}_{\mathrm{MCI}}.$$

Why adjust the P values?

Testing many results increases the chance of a false-positive finding. Benjamini–Hochberg adjustment accounts for these multiple tests; adjusted $P < 0.05$ is treated as evidence of a difference.

NC: frequency-wise MISO versus SISO gain

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_frequency_model_nc_gain.png`

Takeaway: NC MAP gain changed relatively little, whereas CO_2 gain differed across contiguous mid- and higher-frequency regions.

MCI: frequency-wise MISO versus SISO gain

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_frequency_model_mci_gain.png`

Takeaway: The MCI CO_2 pathway showed a broad model effect; MAP gain differences were smaller and concentrated mainly at the lowest frequencies.

Primary band-level evidence for the model effect

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_model_comparison.png`

Takeaway: MISO MAP gain was lower in VVLF in both groups, while the larger and more consistent model-related changes occurred in CO_2 gain.

NC versus MCI: band-level comparisons

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_group_comparison.png`

Takeaway: None of the tested NC–MCI band differences had an adjusted $P < 0.05$. The current data therefore did not identify a robust group difference in these MISO

Conclusions and next steps

1. **Model comparison:** MISO and SISO produced statistically different pathway-specific estimates in several comparisons, particularly for CO_2 gain.
2. **Group comparison:** The tested MISO metrics did not identify a robust difference between NC and MCI in the current data.
3. **Interpretation:** The results support considering multiple measured inputs when estimating autoregulatory dynamics, but adding CO_2 alone was not sufficient to establish an NC–MCI marker. Additional inputs, smaller effects, and biological variability remain possible explanations requiring further investigation.

Next steps

- Obtain longer recordings that permit meaningful cross-validation of SISO and MISO predictions.
- Complete the manuscript and prepare it for journal submission or preprint release.

Backup: frequency-band definitions

Band	Frequency interval	Included frequency bins
VVLF	$0.005 \leq f < 0.024$ Hz	0.0078–0.0234 Hz
VLF	$0.024 \leq f < 0.070$ Hz	0.0313–0.0625 Hz
LF	$0.070 \leq f < 0.200$ Hz	0.0703–0.1953 Hz
HF	$0.200 \leq f \leq 0.350$ Hz	0.2031–0.3438 Hz

Frequency resolution:

$$\Delta f = \frac{4\ \mathrm{Hz}}{512} = 0.0078125\ \mathrm{Hz}.$$

Band averages were calculated within subject first, so each subject contributed one value per band to the formal band-level tests.

Backup: SISO and MISO derivation

For MAP as one input,

$$\mathrm{CBFV} = H_{\mathrm{MAP}}\,\mathrm{MAP} + E, \qquad S_{\mathrm{MAP,CBFV}} = H_{\mathrm{MAP}} S_{\mathrm{MAP,MAP}}, \qquad H_{\mathrm{MAP}} = \frac{S_{\mathrm{MAP,CBFV}}}{S_{\mathrm{MAP,MAP}}}.$$

With MAP and CO_2 entered simultaneously,

$$\mathrm{CBFV} = H_{\mathrm{MAP}}\,\mathrm{MAP} + H_{CO_2}\,CO_2 + E,$$

$$\underbrace{\begin{bmatrix} S_{\mathrm{MAP,MAP}} & S_{\mathrm{MAP},CO_2} \\ S_{CO_2,\mathrm{MAP}} & S_{CO_2,CO_2} \end{bmatrix}}_{\mathbf{S}_{\mathbf{inputs}}} \underbrace{\begin{bmatrix} H^{\mathrm{MISO}}_{\mathrm{MAP}} \\ H^{\mathrm{MISO}}_{CO_2} \end{bmatrix}}_{\mathbf{H}} = \underbrace{\begin{bmatrix} S_{\mathrm{MAP,CBFV}} \\ S_{CO_2,\mathrm{CBFV}} \end{bmatrix}}_{\mathbf{S}_{\mathbf{input,CBFV}}}$$

The implementation solves this spectral system at each frequency using MATLAB matrix left division. The current analysis is unregularized.

Backup: how phase was summarized

Circular mean

$$C = \frac{1}{n}\sum_i \cos\theta_i, \qquad S = \frac{1}{n}\sum_i \sin\theta_i$$

$$\overline{\theta} = \operatorname{atan2}(S, C)$$

Circular spread

$$R = \sqrt{C^2 + S^2}$$

$$s_{\mathrm{circ}} = \sqrt{-2\ln R}$$

Statistics used wrapped phase. For display, the circular group mean was unwrapped across frequency and retained the same circular SD.

Backup: what a BH-adjusted P value means

One raw P value

Tests one null hypothesis without accounting for the other tests in the family.

BH-adjusted P value

Accounts for the number and ranking of related P values. If $P_{\mathrm{BH}} < 0.05$, that test is significant while controlling the false discovery rate within the defined family.

- $P_{\mathrm{BH}} > 0.05$ does **not** prove that two curves or groups are the same.
- More subjects can improve power if the underlying effect is consistent, but sample size does not guarantee significance.
- Standard deviation affects P values because it affects the uncertainty of the estimated difference.

Backup: NC frequency-wise phase comparison

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_frequency_model_nc_phase.png`

Takeaway: Phase comparisons use circular paired differences and permutation tests;

Backup: MCI frequency-wise phase comparison

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_frequency_model_mci_phase.png`

Takeaway: The same circular procedure is used in MCI, with BH adjustment

Backup: NC–MCI frequency-wise gain

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_frequency_group_gain.png`

Takeaway: The isolated adjusted result at 0.1484 Hz is exploratory and was not

Backup: NC–MCI frequency-wise coherence

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_frequency_group_coherence.png`

Takeaway: Frequency-wise group coherence comparisons are exploratory; primary

Backup: NC–MCI frequency-wise phase

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_frequency_group_phase.png`

Takeaway: A high adjusted P value means the available data do not provide evidence

Backup: input coherence and gain change

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_input_association.png`

Takeaway: Several low-frequency associations were significant, but they are exploratory because both variables come from the same spectral matrix.

Backup: MAP–CO_2 pathway balance

FIGURE PLACEHOLDER

Upload this file beside the presentation, into `figures/`, or into `Batch_Figures/`:

`statistical_pathway_balance.png`

Takeaway: MAP partial coherence became larger than CO_2 partial coherence above VVLF, with no corrected NC–MCI difference in balance.

Backup: numerical conditioning

Current observations

- Median HF condition number: 10.9 NC, 15.4 MCI
- HF values > 100: 60/437 NC, 126/988 MCI
- Maximum: 7,467 NC, 17,366 MCI
- High values often occurred with low CO_2 power or large input-power imbalance

Interpretation

- The raw condition number is scale-dependent.
- A high value does not automatically prove physiological collinearity.
- It does indicate that some estimates may be sensitive to small spectral changes.
- Ridge regularization remains a planned sensitivity decision, not part of the reported solve.

Backup: key numeric model comparisons

Group	Path	Band	MISO	SISO	Difference	BH P
NC	MAP	VVLF	1.092	1.332	−0.240	0.0052
MCI	MAP	VVLF	0.837	1.020	−0.183	0.0038
NC	MAP	LF	2.111	2.046	+0.065	0.0464
NC	CO_2	LF	1.761	2.498	−0.737	0.00052
NC	CO_2	HF	2.695	4.944	−2.250	0.00017
MCI	CO_2	VLF	2.066	2.850	−0.784	0.000083
MCI	CO_2	LF	1.599	2.892	−1.293	0.0000021
MCI	CO_2	HF	2.823	5.458	−2.635	0.0052

Gain units are percentage points of baseline CBFV per mmHg. Differences are MISO minus SISO. BH adjustment covered the 16 primary gain tests.